

Aragen Project Report

Mayuri Borkar

1. Data Collection

Ligand activity data for Acetylcholinesterase (AChE) was retrieved from the ChEMBL database using its UniProt ID (P22303). The dataset comprises bioactivity values — including IC50, Ki, Kd, and pChEMBL — for molecules experimentally tested against AChE. Only ligands with relevant activity measurements were retained to ensure data quality.

Molecular structures were extracted in SMILES format via ChEMBL's molecule endpoint. The final curated dataset, **AChE_ligands.csv**, contains ChEMBL IDs, activity values, and SMILES strings for downstream analysis.

2. Data Cleaning and Preprocessing

The dataset underwent a thorough cleaning and preprocessing pipeline to ensure consistency and analytical integrity. Redundant columns (type, units, value) were removed, and numeric fields (pChEMBL value, standard value) were coerced into proper numerical format.

Activity values were normalized to a [0, 1] scale. An **Activity Score** was then derived using a weighted formula — 70% pChEMBL value and 30% standard value — to reflect the relative importance of each metric. Molecules were subsequently classified as **Active** or **Inactive** using a threshold of 0.5. The preprocessed and labeled dataset was saved for further analysis.

3. Feature Extraction

Key molecular descriptors were computed from SMILES representations using the RDKit cheminformatics toolkit. The following physicochemical properties were extracted for each molecule:

- Molecular Weight (MolWt)
- Lipophilicity (LogP)
- Topological Polar Surface Area (TPSA)
- Hydrogen Bond Acceptors (HBA)
- Hydrogen Bond Donors (HBD)
- Rotatable Bonds (RotBonds)

Molecules with invalid SMILES were removed prior to feature extraction. The resulting dataset, **dataset_with_descriptors.csv**, was saved for model training.

4. Model Training and Evaluation

A comprehensive set of machine learning and deep learning models was trained to classify molecules as Active or Inactive against AChE. Models evaluated include Random Forest, XGBoost, Support Vector Machine (SVM), and the transformer-based ChemBERTa architecture.

- **Feature Engineering:** Molecular descriptors and Morgan fingerprints were used as combined input features.
- **Training Protocol:** All models were trained on an 80/20 train-test split, with feature standardization applied where required.
- **Evaluation Metrics:** Performance was assessed using accuracy, precision, recall, and F1-score.

XGBoost with SMILES-based fingerprints combined with molecular descriptors achieved the highest classification accuracy and was selected as the best-performing model.

Class	Precision	Recall	F1-Score	Support
0 (Inactive)	0.84	0.88	0.86	764
1 (Active)	0.88	0.83	0.85	745

Overall XGBoost Accuracy: 85.88%

5. SMILES Generation Using Variational Autoencoder (VAE)

A Variational Autoencoder (VAE) was developed to generate novel drug-like molecules based on the structural patterns of known active AChE ligands. The model operates on character-level SMILES sequences, encoding molecules into a continuous latent space and decoding sampled vectors to produce new molecular structures.

- **Preprocessing:** Active SMILES were tokenized and character-to-index mappings were constructed.
- **Model Architecture:** A bidirectional GRU-based VAE was employed to capture sequential molecular patterns.
- **Training:** The model was trained for 100 epochs, with training loss decreasing from ~120 to ~20, indicating effective learning.

- **Generation:** Novel SMILES were sampled from the learned latent space.
- **Validation:** Chemical validity of generated molecules was assessed using RDKit; invalid SMILES were filtered out.

The training loss curve showed consistent convergence across all 100 epochs, confirming stable model learning.

6. Activity Prediction for Generated Molecules

To assess the biological relevance of generated molecules, the pre-trained XGBoost classifier was applied to predict whether each generated molecule is Active or Inactive. Generated SMILES were converted into Morgan fingerprints (ECFP4, 2048-bit) and passed through the classifier. Each prediction was accompanied by an activity label and a confidence score. Results were stored for downstream filtering and prioritization.

7. Structural Similarity Analysis

Tanimoto similarity scores were computed between newly generated ligands and the pool of known active AChE ligands, using Morgan fingerprints (ECFP4, 2048-bit). For each generated molecule, the maximum similarity score relative to any known active ligand was recorded.

An **average Tanimoto similarity score of 0.8556** was obtained, indicating that the generated ligands are structurally related to known actives — a desirable property for drug discovery applications. Detailed similarity scores were saved in **Similarity_Scores.csv**.

8. Strengths of the Approach

- **High Active Molecule Yield:** The majority of generated SMILES were classified as active, directly supporting drug discovery objectives.
- **Top Classification Performance:** XGBoost outperformed all other trained models, providing reliable activity predictions.
- **Chemically Constrained Generation:** The VAE's character-level vocabulary constrains generation to known chemical substructures, reducing unrealistic outputs.
- **Reliable Data Source:** ChEMBL provides standardized, high-quality bioactivity data, minimizing noise in training.
- **Robust Activity Scoring:** The weighted activity score formula improves discrimination between active and inactive compounds.

9. Limitations and Future Directions

- **Limited Dataset Size:** Only ~7,500 data points from a single protein target constrain the model's generalization capacity.
- **Invalid SMILES Generation:** A fraction of generated molecules are chemically invalid; the pipeline filters these, but the underlying VAE could be further improved.
- **Structural Redundancy:** Approximately 45% of valid generated SMILES closely resemble existing known active molecules, likely a consequence of the small training set limiting structural novelty.

Future work may include expanding the dataset to multiple protein targets, incorporating reinforcement learning for goal-directed molecular generation, and applying 3D docking simulations for further validation of generated candidates.

10. Results — Sample Generated Ligands

A representative subset of generated ligands and their predicted activity classifications (with confidence scores) is presented below:

SMILES	Activity	Confidence
<chem>CC1=Cc2cc(OC(=O)NC3CCN(Cc4ccccc4)CC3)ccc21</chem>	Active	0.54
<chem>Nc1c2c(nc3ccccc13)CCCC2</chem>	Active	0.92
<chem>COC(=O)c1ccccc1CCCCCCCNc1c2c(nc3ccccc13)CCCC2</chem>	Active	0.95
<chem>CCN(CC)C(=O)c1ccc2c(c1)nc(NC(N)=O)cc1n2CCC(C)C</chem>	Inactive	0.94
<chem>C0c1cc2cc(CC(=O)NCCC3CCN(Cc4ccccc4)CC3)c(=O)oc2cc10</chem>	Active	0.86
<chem>CNC(=O)Oc1ccc2nc3n(c(=O)c2c1)CCC3</chem>	Inactive	0.83
<chem>COC(=O)NC1CCC2=CC(OC(=O)N(C)C)C=CC21</chem>	Inactive	0.64
<chem>CNC(=O)c1ccc2c(N3CCN(Cc4ccccc4)CC3)sc2cc10</chem>	Active	0.70
<chem>CN(C)C(=O)Oc1cccc(C(=O)N(C)C)c1</chem>	Inactive	0.56
<chem>CNC(=O)Oc1ccc2c(c1)[C@@]1(C)CCN(C)[C@@H]1N2C</chem>	Active	0.82